

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

November 2013

EE202 Electrical Power Generation

Date: 23.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write the functions of the followings: [05]
 (1) Magnetic separator (2) Electrostatic precipitator
 (3) Screening grid (4) Chemical feeding pipe (5) Condenser
 (b) Draw the layout of Coal Handling Plant. [05]
- Q - 2 (a) With neat sketch explain the layout of thermal power plant in detail. [08]
 (b) Write down the advantages and disadvantages of thermal power plant. [04]

OR

- Q - 2 (a) With neat sketch explain the layout of closed cycle gas turbine power plant. Also write the advantages and disadvantages of gas turbine power plant [08]
 (b) Draw the layout of combined cycle gas turbine power plant. [04]
- Q - 3 (a) Discuss the factors considered for selecting the site to construct a hydro power plant. Also give the advantages and disadvantages of hydro power plant. [06]
 (b) With neat sketch explain the layout of hydro power plant in detail. [07]

OR

- Q - 3 (a) Write down the differences between Pelton wheel turbine and Francis turbine. [07]
 (b) Write down the differences between Kaplan turbine and Francis turbine. [06]

SECTION - II

- Q - 4 (a) Explain the principle of MHD generator and derive the power output equation of MHD generator. [06]
 (b) Write down the differences between open cycle & closed cycle MHD system. [03]
- Q - 5 (a) Give the advantages and disadvantages of nuclear power plant. [03]
 (b) Sketch the nuclear reactor and discuss the different components of nuclear reactor. [05]

OR

- Q - 5 (a) Explain the pressurized water reactor in detail. [05]
 (b) Enlist the different types of fuel cell. Also write the chemical reactions seen in the different types of fuel cell. [03]

- Q - 6 Attempt any three: [18]
 (a) Draw the layout of Diesel engine power plant
 (b) Explain liquid-dominated geothermal power plant in detail.
 (c) How is wind created? Also explain the wind turbine in detail and give the advantages and disadvantages of wind energy.
 (d) Explain different types of solar collectors used for electricity generation.
 (e) Explain Hybrid cycle OTEC with neat sketch and give the advantages and disadvantages of OTEC.